

Selective Encryption for Consumer Applications

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ABSTRACT

Selective encryption is a technique to save computational complexity or enable interesting new system functionality by only encrypting a portion of a compressed bitstream while still achieving adequate security. Although suggested in a number of specific cases, selective encryption could be much more widely used in consumer electronic applications ranging from mobile multimedia terminals through digital cameras were it subjected to a more thorough security analysis. We describe selective encryption and develop a simple scalar quantizer example to demonstrate the power of the concept, list a number of potential consumer electronics applications, and then describe an appropriate method for developing and analyzing selective encryption for particular compression algorithms. We summarize results from application of this method to the MPEG-2 video compression algorithm.

INTRODUCTION

Claude Shannon first pointed out (in one of his classic 1940s papers [1]) the strong relationship between data compression and encryption. In particular, he showed how the redundancy in a source (e.g., the redundancy in the English language associated with the different relative frequencies of letters and letter groupings) makes encryption weak: an attacker can use the different frequencies in symbols in the source material to guess the encryption mapping from source material to encrypted material. Surprisingly, for the English language and a simple substitution cipher in which each letter is mapped to some other letter, the mapping is close to uniquely determined and the cipher broken after observing about 30 letters of the ciphertext. He then suggested that data compression, by removing the redundancy in the source, could strengthen encryption. Perfect compression, in fact, would eliminate any statistical redundancy in the source to the encryptor, and an attacker could do no better than successively guess possible encryption mappings (the mappings being indexed by

the encryption key shared through some other means by the encryptor and the decryptor).

Inspired by the way compression can strengthen encryption, system designers have recently been asking a different but related question: can we secure a compressed data stream if only part of it is encrypted? This is the concept of *selective encryption* shown in Fig. 1. For selective encryption to work, we need to rely not only on the beneficial effects of redundancy reduction described by Shannon, but also on a characteristic of many compression algorithms to concentrate important data about reconstruction in a relatively small fraction of the compressed bitstream. These important elements of compressed data become candidates for selective encryption; without them, an attacker can make little use of the rest of the compressed bitstream even if it is in the clear.

But, why bother? What is the advantage of only encrypting part of a compressed bitstream? There are two answers, one fairly obvious and the other less so.

The more obvious answer is that selective encryption reduces the total computational complexity of processing a signal that needs to be both compressed and encrypted by the simple expedient of saving cycles that would be spent on encryption. The combination of compression and encryption is common and likely to become more so. In consumer applications, essentially all digital media sources (speech, images, audio, video) are compressed as a matter of course when delivered over networks and, increasingly, encrypted either for reasons of privacy or to protect the rights of a content owner. And as we work to make access to digital content ubiquitous, we often want to use devices in which space or power limitations make savings in computation valuable.

The less obvious answer is that selective encryption can enable new types of system functionality. One intriguing consumer electronics application is the efficient overlay of more than one security system on a single compressed bitstream. Imagine, for example, that you are operating a digital cable television system using an open standard compression format (e.g., the MPEG-2 format used in most cable and satellite

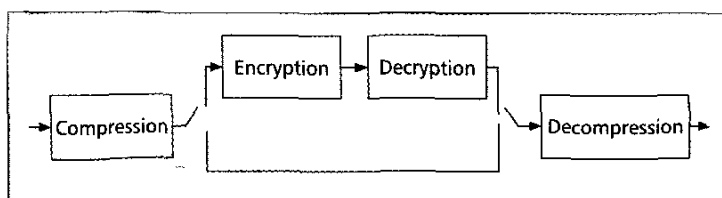
digital television systems), but with an encryption system that is proprietary to your equipment supplier. You want to consider an alternative supplier, but it would be extraordinarily expensive to replace all of your current equipment (particularly the large quantities of set-top boxes in consumer homes) to enable a different security system, and you do not want to spend the bandwidth necessary to duplicate every channel you send using both the old and new suppliers' security systems so as to enable old and new set-top boxes to work on the same network. Selective encryption gives you another option: only encrypt a small portion, say a few percent, of each channel and do so using the security systems of each of your suppliers. Send the rest of the channel's content once in the clear. Now, for a few percent bandwidth overhead, you have enabled both new and old suppliers on the same network.

The example of multiple security system overlay is not just theoretical; the newly proposed Passage system from Sony offered to the U.S. cable industry, currently in field trials, is based on this exact principle [2]. Other examples of novel system functionality enabled by selective encryption are developed later.

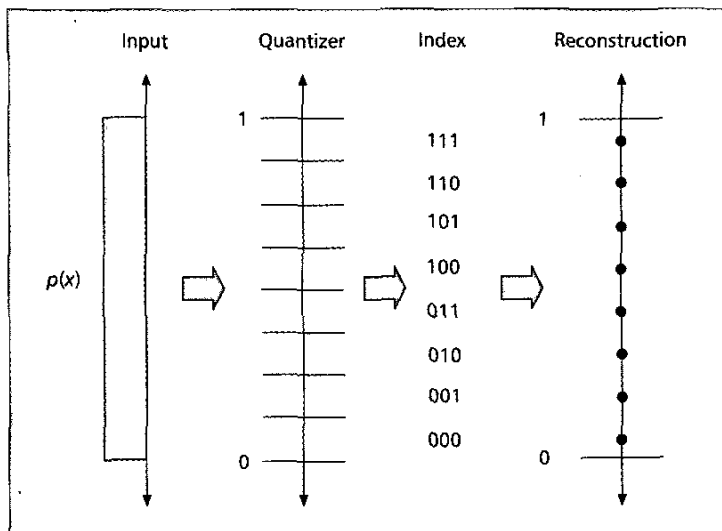
While selective encryption opens up the possibility of computational savings and new system functionality, it also comes with some important challenges. In particular, intermixing encryption and compression is an example of a layer violation. We often conceive and implement communication systems with the idea of layers of processing that are independent of each other, at least to the extent that the internal operation of each layer is irrelevant to layers above or below. The reduction in system complexity facilitated by this partitioning is not to be taken lightly, and mixing the internal operations of traditionally separate layers such as compression and encryption needs to be motivated by nontrivial performance advantages. The second challenge of selective encryption is the traditional challenge of security systems: how do you show a system is secure? With the exception of provable mathematical properties of certain basic encryption algorithms, almost all security systems are not provably secure in the general sense. Rather, confidence in security is based on a plausibility argument: how plausibly have advocates of a particularly security system evaluated the range of possible attacks and the safety of the proposed system from each one? From the security researchers' perspective, it is not sufficient to show that a system is secure against a naïve or casual user; we must also show that it is secure against attackers at least as clever and resourceful as we are.

A SCALAR QUANTIZER EXAMPLE AND SHANNON AGAIN

Before considering the range of possible applications of selective encryption, particularly in consumer devices and networking, we can get a good feel for the potency of the approach by examining in detail a particular common, simple, and analytically tractable system: a scalar quantizer (a basic fraction of the ubiquitous analog-



■ **Figure 1.** Selective encryption. The choice of which parts of the compressed bitstream to encrypt is made so as to render the unencrypted portion useless to an eavesdropper.



■ **Figure 2.** A simple 3-bit scalar quantizer operating on a uniformly distributed input over the range 0 to 1.

digital converter). Our key question is: how much and what part of the bitstream produced by such a quantizer do we need to encrypt to make it secure against an attacker — an attacker who can use all the information we leave in the clear but cannot break the encryption we have applied to the portion we have chosen to encrypt? Do we need to encrypt all or nearly all the information to make it secure, or can we get away with only encrypting a small fraction?

We consider the almost trivial example of a uniform 3-bit scalar quantizer operating on a uniformly distributed continuous valued input on the range 0 to 1, drawn from a memoryless source, one for which each sample is independent. This very simple source is not representative of any particular real-world source, but allows us to understand the basic power of selective encryption. The operation of this quantizer, coding the input into one of eight possible reproductions at the receiver, is shown in Fig. 2. In the absence of any encryption, this quantizer has a mean square error of 0.001302, calculated by taking the average square error resulting from replacing an input within a bin with a reconstruction at the center of a bin.

Selective encryption involves scrambling a particular subset of the bits in the transmitted bin indices. Since we assume the attacker cannot break the encryption applied to the bits we scramble, the best she can do is guess those bits

Subset of bits encrypted	Fraction of bits encrypted	Mean square error
$\emptyset$	0.0	0.0013
{0}	0.3333	0.0091
{1}	0.3333	0.0325
{2}	0.3333	0.1263
{0,1}	0.6666	0.0403
{1,2}	0.6666	0.1575
{0,2}	0.6666	0.1341
{0,1,2}	1.0	0.1654

■ **Table 1.** Fraction of bits encrypted and mean square error as a function of which subset of bits in the quantizer bit index to encrypt.

(being right half the time and wrong the other half). With bit 2 as the most significant bit and bit 0 as the least significant, we can enumerate all possible choices of bits to scramble and the resulting distortion suffered by the attacker, shown in Table 1. It's obvious from this table what the optimal strategy is for a given fraction of bits encrypted: if we want to scramble 1/3 of the bits, we should use the set {2}, if we want to scramble 2/3 of the bits, we should use the set {1,2}. This should surprise no one; there is a good reason for calling the most significant bit *most significant*. More interestingly, we graph selective encryption performance in Fig. 3. This is encouraging as it shows graphically the hoped for distortion curve shape; we can get most of the distortion gain (and it is sizeable) quickly as we increase the fraction of bits encrypted.

An even more encouraging insight comes from considering an attacker's default alternative: ignoring the received bitstream altogether. If an attacker did this, she would simply choose to reproduce all values with the center of the range from 0 to 1 (she would act as if she was using a 0-bit quantizer); her resulting mean square error would be 1/12 or 0.08333. But we generate distortion greater than this even when we are scrambling only 1/3 of the bits. In other words, by scrambling just 1/3 of the bits, we have made the received stream worse than useless to the attacker; she might as well guess a default reconstruction value and ignore the stream altogether. Thus, our original question — do we only need to scramble a small fraction of the bits to render the stream secure — is answered positively.

Finally, we can adopt a security analyst's perspective by asking how we might attack such a system. Can we deduce anything about a scrambled bit from its underlying distribution or from other bits (that are in the clear)? Assuming the input is uniformly distributed across the whole range and independent from sample to sample, each bit in the quantized index will be statistically independent of all other bits and take on the value both 0 and 1 with equal probability; we cannot guess anything useful about the encrypted bits by themselves or from the values of in-the-clear bits. So our cryptanalysis exposes no weakness from selective encryption per se (the strength of the system will be solely determined

by the cryptographic strength of the algorithm applied to the encrypted bits themselves).

This last property — that the in-the-clear portion of the stream is statistically independent of the scrambled portion and so does not help us at all in guessing what has been hidden — can be seen as a property of a compression system that is "perfect" in the sense that it has reached the limits on compression outlined by Shannon. A compression system that has reached this rate distortion limit must necessarily have no statistical dependence left in the compressed bitstream. Why not? If it did, we could design a further stage of compression that would exploit the statistical dependence and achieve an additional improvement in compression; if this were possible, the original system must not have been perfect. With no statistical dependency left, there is nothing useful to be guessed about the encrypted bits from the in-the-clear bits. This is a further encouraging property of compression and selective encryption (not just the case of encrypting all the bitstream as considered by Shannon), but our enthusiasm needs be tempered by the fact that most practical compression systems do not achieve perfection.

APPLICATION OF SELECTIVE ENCRYPTION TO CONSUMER APPLICATIONS

Since compression (particularly of various types of digital media content) and encryption occur together frequently in consumer electronics, it is not hard to find a number of cases where selective encryption might apply. Following are a set of examples, some already examined by previous researchers and some new, suggestive of the range of such applications.

MOBILE MULTIMEDIA TERMINALS AND HOME MULTIMEDIA NETWORKING

Mobile multimedia terminals are emerging from both cell phone and personal digital assistant (PDA) heritages. Such terminals increasingly include low to moderate resolution color displays, cameras, and moderate native processing power. The ability to send and receive voice, music, pictures, and video is becoming commonplace. However, battery size and battery life is almost always a concern for these devices. Selective encryption can be used to reduce the power consumed by the encryption function for digital content when, for example, the content is protected by a digital rights management system. Furthermore, it is not necessary to completely obscure the content to support rights management; it simply needs to be rendered undesirable enough that a negligibly small fraction of the population would choose a free but damaged version over paying for an undamaged version.

For the same reason, selective encryption should be a candidate for protecting content in a home multimedia network where some of the receiving devices are expected to be mobile (i.e., battery powered) or meant to be very inexpensive (so that saving computational complexity is very important).

MULTIPLE SECURITY SYSTEMS

It is sometimes commercially convenient to operate more than one security system protecting a single piece of content — the particular example developed in the introduction of this article. This may be necessary in a gradual swap of one system for another (if, e.g., the first one has been breached sufficiently to warrant its replacement but still has some protective value) or to encourage competition between two equipment suppliers, each of which provides its own security system.

Selective encryption allows a much more bandwidth-efficient solution than sending content in parallel under each encryption system; if 5 percent of the material is encrypted, say, the bandwidth requirement is 95 percent (material in the clear that is shared by all systems) plus the number of systems times 5 percent (the independently secured material for each system). An even more elegant solution than selective encryption occurs if a single scrambling is applied to all of the stream, with different security vendors coordinating to each distribute key for the scrambling using their own security systems, as is the case in DVB simulcrypt [3]. But this solution may be precluded (and hence selective encryption attractive) if not all the security system vendors wish to cooperate or they cannot agree on a common scrambling solution.

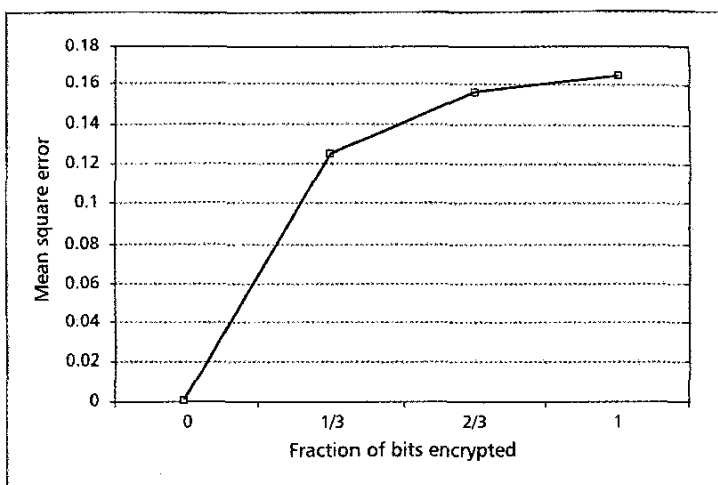
SOFTWARE RENEWABLE SECURITY

The possibility of substantially changing a widely deployed security system has proven invaluable to operators of digital rights management systems attempting to protect high value content (e.g., movies) from pirates. Such systems often envision a sequence of countermeasures to be deployed in a never-ending battle with pirates [4]. Reducing the cost of changing or replacing a rights management system is consequently quite valuable to a system operator or a digital rights management system provider.

Because a selective encryption system can be designed to only encrypt a small part of a bit-stream, it is feasible to move more or potentially all of the security system implementation into software. This can greatly increase options for economically updating security systems for large quantities of devices already deployed in the field.

CONTENT CACHING

Storing bulk content in the clear has the advantage of simplifying content production and reducing the amount of material that can directly be used to attack an encryption algorithm (generally, encryption algorithms are more vulnerable the more encrypted material there is to analyze). In fact, if bulk content does not need to be protected, it is feasible to distribute it quite promiscuously, say, on public peer-to-peer networks or even pre-delivered to a user's location and stored on a general-purpose publicly accessible hard drive. A small portion of the content (the encrypted portion) can be maintained remotely and delivered when it is appropriate to enable use of the content. As with the multiple security systems and renewable software security applications, this also facilitates the effi-



■ **Figure 3.** Mean square error for a selectively encrypted 3-bit uniform scalar quantizer as a function of the fraction of bits encrypted.

cient use of independent security systems and the efficient replacement of security systems.

SAMPLING OF CONTENT

A closely related application to content caching is making available low-quality previews of content that can be converted to full-quality reproductions by adding encrypted content that is then decrypted. Depending on how the selective encryption system is configured, it may be that recognizable but substantially damaged reproductions can be generated without attempting to decode the selectively encrypted portion; this content can be made publicly and widely available. The full quality can be rendered once the key is obtained to decode the selectively encrypted portion.

DIGITAL CAMERAS AND PRIVACY

Most of the preceding examples are suited to digital rights management applications in high-value multimedia distribution. A different application area is in preserving the privacy of material transmitted between individuals. A particular consumer electronics version of this is sourcing pictures using low-cost digital cameras. Such cameras are increasingly enabled with low-cost ways to transfer such images to networks (e.g., various network connections, removable storage media). Consumers may be interested in protecting their pictures from prying eyes. But, digital cameras usually have limited computational resources and power. A technique that allows a consumer to adequately secure his or her picture content while only encrypting a small portion of the image content (say, in JPEG or JPEG2000 compression format) could represent an attractive solution to picture privacy.

SECURING COMMON CONSUMER-ORIENTED APPLICATIONS

While many of the applications of the previous section could be quite interesting, before system operators or content owners are willing to trust their content to selective encryption systems

there needs to be a careful and thorough analysis of the security of such systems. To satisfy the "plausibility" approach of system security analysis, we need to imagine the attacks that a capable and resourceful opponent might use to breach a proposed system, and we need to do so systematically; such attacks may or may not be focused on breaking the encryption algorithm in the system; any weakness is fair game.

A STANDARD CRYPTANALYTICALLY MOTIVATED APPROACH

One good overall approach follows these steps:

- Consider the syntactic elements of a particular compressed bitstream. Determine what fraction of the bitstream they typically represent (either analytically or experimentally by observing representative bitstreams and representative encoders).
- Estimate the impact that having the wrong value for this syntactic element would have on decompression.
- Form a figure of merit as the impact of wrong values divided by the fraction of the bitstream used for each syntactic element.
- Rank syntactic elements in descending order of figure of merit. Consider a family of selective encoders to encrypt the first such element, then the first two such elements, then the first three such elements, and so forth.
- Develop a set of attacks intended to compromise these selective encryption systems. Attacks should include *statistical attacks* based on residual correlation between in the clear and encrypted material, *usage attacks* based on the fact that encoders may be configured statically or in a manner that can be predicted from external information (e.g., the size of the picture may be predictable if it is advertised as an HDTV broadcast), and *perceptual attacks* based on using error concealment strategies to mitigate the effect of missing (scrambled) information (the very same strategies that are used in error concealment to mitigate the effects of channel errors on a reconstruction).
- Evaluate the effectiveness of the attacks, determine feasible counterattacks if possible, and iterate.

MPEG-2 VIDEO EXAMPLE

We have applied this approach to the selective encryption of MPEG-2 video (used in most contemporary digital television applications) in [5]. Some of the key points reported there include:

- Typical high-performance MPEG-2 encoded bitstreams only use a small portion of bits (around 1 percent) in important headers (video sequence, group of pictures, picture, and slice). It can be simple to obscure such headers because of a usual practice in encoding of aligning these headers and the multiplex (transport) level at which encryption is performed.

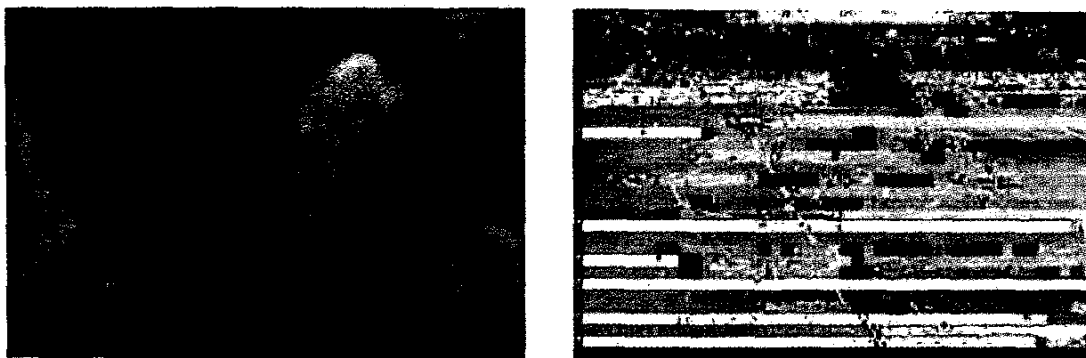
- However, fields in such headers can be quite vulnerable to attack, even if obscured by selective encryption, for a variety of reasons: the fields are often static, they can be guessed from external information that is probably available to an attacker, they can be guessed from other information in the bitstream (e.g., picture type

can be guessed from picture size, an example of the cryptanalytic technique of *traffic analysis*), or they can be ignored, albeit with nontrivial consequences for decoded image quality. We evaluated each of these fields, and proposed and tested attacks. For example, we showed that a perceptual attack on the *quantizer_scale_code* syntactic element is feasible albeit with nontrivial picture degradation: in typical sequences there is a strongly peaked distribution for this code, and a perceptual attack would be to always use an expected value for this code in place of the correct value. The results of an attack are shown in Fig. 4; clearly the resulting reconstruction is distorted, but it is not obvious that it is sufficiently distorted to cause a pirate to pay for a clean version if the distorted version is available for free. A more encouraging example is the choice of the *macroblock_type* field that signals to a decoder the type of prediction used for each macroblock (16-pixel vertical by 16-pixel horizontal region) in a video frame. Although this field does not use a very large fraction of the bitstream (on the order of a couple of percent typically), if absent it is very difficult for a decoder to guess it and to decode remaining material correctly (since the macroblock type uses a Huffman code and, if incorrectly decoded, a decoder has a hard time resynchronizing). Figure 4 shows an example of attempted decoding by an intelligent decoder when just 20 percent of the *macroblock_type* bits have been obscured (meaning well under 1 percent of the total bitstream).

- The effectiveness of MPEG-2 video selective encryption when high performance (in terms of only needing to encrypt a very small portion of the bitstream) is targeted is quite dependent on how a compressor is operated. We distinguish between three ways to operate a compressor: *cooperative*, *neutral*, and *antagonistic*. A cooperative compressor is one that performs compression in a manner likely to enhance the security of a selective encryption scheme, for example, by concentrating important information (so that only a small fraction of bits need be encrypted), and ensuring that this important information itself is hard to guess from outside factors and is not static. A neutral compressor does not make any particular effort to compress in a way to enhance or degrade selective encryption (e.g., it may be optimized for compression alone), and an antagonistic compressor compresses in a manner intended to make it unrewarding to apply selective encryption (e.g., by attempting to make selective encryption ineffectual unless nearly all of the bits are encrypted). A cooperative encoder can, in fact, make secure even selective encryption with a small fraction of bits encrypted. An antagonistic encoder can render selective encryption quite insecure.

CONCLUSIONS

Selective encryption is still an active area of research, in terms of both proposing and validating specific applications (e.g., with particular compression algorithms in mind), and developing a theoretical framework. There are strong parallels between selective encryption and other active areas of development in communication, including:



■ **Figure 4.** A movie frame showing the effect of a perceptual attack on encrypted quantizer scale code (fixed at 5 in the reconstruction) (left) and a sports frame showing the effect of 20 percent encryption of macroblock type (right).

Progressive transmission: Some applications find it particularly valuable to be able to generate a crude but useful approximation of the original signal using only a small portion of the received bitstream, then gradually build up a better and better reproduction as more information is received. A simple example is browsing images over a low-bandwidth communication link; an initial cheap-to-transmit crude reproduction may be good enough to reject a particular image as uninteresting, with only a desirable image being allowed to continue to complete high-fidelity reconstruction. Designing a good progressive transmission system produces good candidates for selective encryption, since the notion of concentrating important information to enable initial good reproductions is related to the notion of concentrated information that, if encrypted, makes the remaining information in a compressed bitstream much less useful. And, as in the case of perfect compression, there are encouraging information theoretic arguments for selective encryption security in the case of perfect progressive transmission systems.

Joint source-channel coding: Researchers have considered the problem of how to allocate the error protection available in channel coding optimally across the different elements of a source coding (compression) scheme. The idea of determining which portions of a compressed format are most important to producing a good reproduction (for the purpose of protecting them) is analogous to the problem in selective encryption of determining which portions of a compressed format are most important to a good reproduction (for the purpose of obscuring them).

Selective encryption is not widely used yet, in spite of the fact that it offers particular and interesting complexity, cost, and system functionality advantages. We suspect that there are two key reasons for this: first, although selective encryption has been proposed in some specific applications, the breadth of its applicability has not yet been widely understood (nor has the correspondingly broad theoretical base yet been developed), and second, the majority of investigations to date have not been conducted in a manner to inspire confidence in security experts.

As researchers address this issues, selective encryption could see widespread application wherever compression and security occur together and where cost, complexity, or power are at a premium or the other system innovations we've discussed would be helpful.

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